

Replacing ngrok with Cloudflare Tunnel for n8n

This documents the exact steps taken to move the n8n Docker setup off ngrok (1 GB/month + 20,000 request limits) onto Cloudflare Tunnel (no bandwidth or request caps).

Project values used:

- Domain: `sidshkya.com`
 - Subdomain for n8n: `n8n.sidshkya.com`
 - Tunnel name: `n8n-tunnel`
 - Tunnel ID: `c30f5aac-36d0-49db-90cf-a2097035c05a`
 - Host: Ubuntu Linux, Docker Compose
 - Project folder: `~/Desktop/n8n-automation`
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1. Buy a domain on Cloudflare Registrar

Cloudflare Tunnel needs a stable hostname, which means a domain managed by Cloudflare's DNS (a free `trycloudflare.com` URL changes on every restart, which breaks OAuth redirect URIs).

1. Go to `dash.cloudflare.com` and log in / sign up (free).
2. Open **Domain Registration** in the dashboard.
3. Search for a name (not a TLD — e.g. search `sidshkya`, not `com`).
4. Pick an available option and register it. Buying directly through Cloudflare Registrar means the domain is added to your account with Cloudflare as the DNS provider automatically — no manual nameserver switch or propagation wait.
5. Confirm the domain shows **Active** in your dashboard.

⚠ Domain registration is an **annual renewal fee**, not a one-time purchase — e.g. `sidshkya.com` renews at ~\$10.46/year. Cloudflare charges at-cost with no renewal markup, but it's not "pay once, own forever."

2. Install and log in to `cloudflared` (via Docker)

No need to install `cloudflared` natively — the official Docker image works directly, keeping things consistent with the rest of the stack.

```
cd ~/Desktop/n8n-automation
mkdir -p cloudflared
docker run -it --rm -v ./cloudflared:/home/nonroot/.cloudflared
cloudflare/cloudflared:latest tunnel login
```

Important: the `cloudflared` image runs as a non-root user (`nonroot`), whose home directory is `/home/nonroot`, **not** `/root`. Mounting to `/root/.cloudflared` silently fails to persist anything — always mount to `/home/nonroot/.cloudflared`.

This prints a URL — open it in a browser, log into Cloudflare, and authorize against your domain. It downloads `cert.pem` into `./cloudflared` on success.

3. Create the tunnel

```
docker run -it --rm -v ./cloudflared:/home/nonroot/.cloudflared
cloudflare/cloudflared:latest tunnel create n8n-tunnel
```

This prints a **Tunnel ID** and writes a credentials file named `<tunnel-id>.json` into `./cloudflared`. Both `cert.pem` and the credentials JSON should now be in that folder.

4. Route a hostname to the tunnel

```
docker run -it --rm -v ./cloudflared:/home/nonroot/.cloudflared
cloudflare/cloudflared:latest tunnel route dns n8n-tunnel n8n.sidshkya.com
```

Run this from the project root (`~/Desktop/n8n-automation`), not from inside the `cloudflared` subfolder — running it one level too deep silently breaks the relative `./cloudflared` mount path and causes a "Cannot determine default origin certificate path" error.

Double-check the domain spelling. A single mistyped character here (`sidshky.com` instead of `sidshkya.com`) creates a broken nested CNAME record like `n8n.sidshky.com.sidshkya.com`, which doesn't route correctly. If this happens: delete the bad record from the dashboard (Section 6) and re-run the command with the correct domain.

This creates a CNAME record automatically in your Cloudflare DNS — no manual DNS dashboard editing needed for a correct run.

5. Write the config files

cloudflared/config.yml:

```
tunnel: c30f5aac-36d0-49db-90cf-a2097035c05a
credentials-file: /home/nonroot/.cloudflared/c30f5aac-36d0-49db-90cf-
a2097035c05a.json

ingress:
  - hostname: n8n.sidshkya.com
    service: http://n8n:5678
  - service: http_status:404
```

docker-compose.yml (ngrok service replaced with `cloudflared`):

```

services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    container_name: n8n
    restart: unless-stopped
    ports:
      - "5678:5678"
    environment:
      - N8N_HOST=n8n.sidshkya.com
      - N8N_PROTOCOL=https
      - WEBHOOK_URL=https://n8n.sidshkya.com/
      - N8N_PORT=5678
      - N8N_LICENSE_ACTIVATION_KEY=${N8N_LICENSE_ACTIVATION_KEY}
      - N8N_PROXY_HOPS=1
      - N8N_ENABLED_MODULES=instance-ai
      - N8N_INSTANCE_AI_MODEL=openrouter/${OPENROUTER_MODEL}
      - N8N_INSTANCE_AI_MODEL_API_KEY=${OPENROUTER_API_KEY}
      - N8N_INSTANCE_AI_SANDBOX_ENABLED=true
      - N8N_INSTANCE_AI_SANDBOX_PROVIDER=daytona
      - N8N_INSTANCE_AI_SANDBOX_IMAGE=daytonaio/sandbox:0.5.3-slim
      - DAYTONA_API_URL=https://app.daytona.io/api
      - DAYTONA_API_KEY=${DAYTONA_API_KEY}
      - INSTANCE_AI_BRAVE_SEARCH_API_KEY=${BRAVE_SEARCH_API_KEY}
    volumes:
      - n8n_data:/home/node/.n8n
  dns:
    - 8.8.8.8
    - 1.1.1.1

  cloudflared:
    image: cloudflare/cloudflared:latest
    container_name: n8n-tunnel
    restart: unless-stopped
    depends_on:
      - n8n
    command: tunnel --config /home/nonroot/.cloudflared/config.yml run
    volumes:
      - ./cloudflared:/home/nonroot/.cloudflared:ro

volumes:
  n8n_data:
    external: true

```

Remove the old `NGROK_AUTHTOKEN` and `NGROK_DOMAIN` variables from `.env` — no longer needed.

6. Clean up any bad DNS records

If a typo created a broken record (e.g. `n8n.sidshky.com`), delete it manually:

1. dash.cloudflare.com → select the `sidshkya.com` zone.

2. Open **DNS** in the left sidebar.
3. Find the incorrect CNAME record (check for missing/extra letters).
4. Click the delete icon on that row and confirm.
5. Leave the correct **n8n.sidshkya.com** record untouched.

7. Update the Google OAuth redirect URI

In [Google Cloud Console](#) → your OAuth client → **Authorized redirect URIs**:

- Remove the old ngrok URL.
- Add: **<https://n8n.sidshkya.com/rest/oauth2-credential/callback>**

Skipping this step causes Gmail/Sheets OAuth to fail silently after the switch.

8. Bring the stack up and verify

```
docker compose down
docker compose up -d
docker logs n8n-tunnel
```

Check the logs for a successful tunnel connection, then visit **<https://n8n.sidshkya.com>** to confirm the editor loads, and re-authenticate Gmail/Google Sheets to confirm the new OAuth callback works.